

Experiment 4

Experiment to understand data preprocessing in Data Science.

Code 1

```
import numpy as np
import pandas as pd

df = pd.read_csv("pre-process_datasample.csv")
print("Original Dataset:")
print(df)
print()

print("Dataset Info:")
df.info()
print()

print("Mode of Country column:")
print(df.Country.mode())
print()

df['Country'] = df.Country.fillna(df.Country.mode()[0])
df['Age'] = df.Age.fillna(df.Age.median())
df['Salary'] = df.Salary.fillna(round(df.Salary.mean()))
print("Dataset after handling missing values:")
print(df)
print()

print("One-Hot Encoded Country column:")
print(pd.get_dummies(df.Country))
print()

updated_dataset = pd.concat([pd.get_dummies(df.Country), df.iloc[:, [1, 2, 3]]],
axis=1)
print("Updated dataset after One-Hot Encoding:")
print(updated_dataset)
print()
print("Updated Dataset Info:")
updated_dataset.info()
print()

updated_dataset['Purchased'] = updated_dataset.Purchased.replace(['No', 'Yes'], [0,
1])
print("Final Preprocessed Dataset:")
print(updated_dataset)
```

Output

```
Original Dataset:
   Country  Age  Salary  Purchased
0  France  44.0  72000.0         No
1   Spain  27.0  48000.0         Yes
2  Germany  30.0  54000.0         No
3   Spain  38.0  61000.0         No
4  Germany  40.0     NaN         Yes
5  France  35.0  58000.0         Yes
6   Spain  NaN  52000.0         No
7  France  48.0  79000.0         Yes
8  Germany  50.0  83000.0         No
9  France  37.0  67000.0         Yes

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Country    10 non-null      object
1   Age         9 non-null       float64
2   Salary      9 non-null       float64
3   Purchased   10 non-null      object
dtypes: float64(2), object(2)
memory usage: 452.0+ bytes

Mode of Country column:
0    France
```

Name: Country, dtype: object

Dataset after handling missing values:

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	63778.0	Yes
5	France	35.0	58000.0	Yes
6	Spain	38.0	52000.0	No
7	France	48.0	79000.0	Yes
8	Germany	50.0	83000.0	No
9	France	37.0	67000.0	Yes

One-Hot Encoded Country column:

	France	Germany	Spain
0	True	False	False
1	False	False	True
2	False	True	False
3	False	False	True
4	False	True	False
5	True	False	False
6	False	False	True
7	True	False	False
8	False	True	False
9	True	False	False

Updated dataset after One-Hot Encoding:

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	No
1	False	False	True	27.0	48000.0	Yes
2	False	True	False	30.0	54000.0	No
3	False	False	True	38.0	61000.0	No
4	False	True	False	40.0	63778.0	Yes
5	True	False	False	35.0	58000.0	Yes
6	False	False	True	38.0	52000.0	No
7	True	False	False	48.0	79000.0	Yes
8	False	True	False	50.0	83000.0	No
9	True	False	False	37.0	67000.0	Yes

Updated Dataset Info:

```
<class 'pandas.core.frame.DataFrame'>
```

RangeIndex: 10 entries, 0 to 9

Data columns (total 6 columns):

#	Column	Non-Null Count	Dtype
0	France	10 non-null	bool
1	Germany	10 non-null	bool
2	Spain	10 non-null	bool
3	Age	10 non-null	float64
4	Salary	10 non-null	float64
5	Purchased	10 non-null	object

dtypes: bool(3), float64(2), object(1)

memory usage: 402.0+ bytes

Final Preprocessed Dataset:

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	0
1	False	False	True	27.0	48000.0	1
2	False	True	False	30.0	54000.0	0
3	False	False	True	38.0	61000.0	0
4	False	True	False	40.0	63778.0	1
5	True	False	False	35.0	58000.0	1
6	False	False	True	38.0	52000.0	0
7	True	False	False	48.0	79000.0	1
8	False	True	False	50.0	83000.0	0
9	True	False	False	37.0	67000.0	1

Output

```
C:\Users\vsros\AppData\Local\Temp\ipykernel_20308\1860359933.py:40: FutureWarning: Downcasting behavior in `replace` is
  updated_dataset['Purchased'] = updated_dataset.Purchased.replace(['No', 'Yes'], [0,
```